

Justin Dugan

Objective

Computer Engineering student at Georgia Tech with hands-on experience in embedded controls, numerical simulation, and low-level systems programming. Seeking Spring 2027 research and Summer 2027 internship opportunities in robotics, computational science, or applied math.

Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Computer Engineering, Minor in Mathematics, GPA 3.85

Aug 2025 – Present

Expected Graduation, May 2029

Skills

Programming: C++, Java, C#, Python, GLSL

Robotics & Control: State-space control, trajectory generation, odometry, autonomous systems

Scientific Computing: Numerical integration, optimization, FFT

Systems & Tooling: Linux, Docker, Git, Teensy, Raspberry Pi

Experience

RoboJackets (Georgia Tech) | Atlanta, GA

Sep 2025 – Present

RoboRacing Project Manager

Autonomous robotics team competing in Micromouse and other embedded challenges.

- FSM-based motion planning with trapezoidal velocity profiles and feedforward+PID motor control; competed at APEC 2026
- Built peripheral drivers (I²C IMU, ADC-sampled IR wall sensors, PWM motor control via DRV8833) and integrated them into a closed-loop odometry pipeline
- Designed IR-based wall detection and used those readings to correct dead-reckoning odometry between maze cells

FIRST Robotics Competition (Team 1640 “Sab-BOT-age”) | Exton, PA

Sep 2022 – Aug 2025

Lead Programmer, Team Captain, Drive Team

International high school robotics competition

- Sole architect of the 2023–2025 robot control stack, including state-space control, autonomous routines, and vision-based pose estimation and odometry
- Designed a novel swerve drive control algorithm improving driver input-to-module state mapping, reducing oscillation and significantly improving controllability in competition
- Awarded Innovation in Control (World Championship, 2025) and Autonomous Award (World Championship, 2024)

The Robot Collective VIP (Georgia Tech) | Atlanta, GA

Jan 2026 – May 2026

Undergraduate Researcher

- Designed a battery monitoring system for multi-robot platforms
- Trained and optimized a YOLOv5 object detection model for identifying robots in multi-agent environments; exported to ONNX for deployment on Raspberry Pi

Mathnasium | West Chester, PA

Aug 2023 – Present

Instructor

After-school mathematics learning center providing individualized instruction from arithmetic through calculus.

- Provided individualized and small-group math instruction to 5–25 students, adapting to diverse skill levels in real time
- Collaborated with instructors and center leadership to coordinate lesson pacing and instructional strategies

Projects

Fractal Renderer / Expression-to-GLSL Compiler | Personal

May 2025 –

May 2026

Designer & Developer

Renders fractals from arbitrary user-defined math expressions in real time.

- Built a custom expression parser and GLSL code generator to compile user-defined math into GPU shaders at runtime
- Enabled real-time exploration by dynamically recompiling GPU shaders from modified expressions without restart
- Improved numerical stability and precision by implementing a custom complex arithmetic library and hi–lo floating-point representation, increasing accuracy at deep zoom levels

6502 Breadboard Computer | Personal

2024

Designer & Developer

- Designed and assembled a functional 6502-based computer on a breadboard, including clocking, memory, and I/O.
- Debugged timing, signal integrity, and bus contention issues at the hardware level to achieve stable operation.
- Wrote 6502 assembly programs to verify bus timing, memory mapping, and I/O functionality

Physics Simulator & Numerical Optimizer | Personal

2024–2025

Designer & Developer

Charged-particle dynamics simulator with evolutionary trajectory optimization under electromagnetic fields

- Derived and implemented differential equation models for charged-particle motion under magnetic and electrostatic forces
- Designed a differential evolution optimizer evaluating 600k–2M parameter configurations per run to optimize trajectories
- Accelerated simulation throughput using multithreading and JIT compilation, reducing runtime by ~20x